

Rezolvare — Bilet de examen SAE

1. $G(s) = \frac{1}{s(s+5)}, T_e = 0.1s$

a) $H(z)$ – metoda trapezului (Tustin)

$$s = \frac{2}{T_e} \cdot \frac{z-1}{z+1} = 20 \frac{z-1}{z+1} \quad s+5 = \frac{20(z-1)+5(z+1)}{z+1} = \frac{25z-15}{z+1}$$

$$G(z) = \frac{1}{s(s+5)} = \frac{(z+1)^2}{20(z-1)(25z-15)} = \frac{(z+1)^2}{100(z-1)(5z-3)} = \frac{z^2+2z+1}{500z^2-800z+300}$$

$$H(z) = \frac{G(z)}{1+G(z)} = \frac{z^2+2z+1}{(500z^2-800z+300)+(z^2+2z+1)}$$

$$H(z) = \frac{z^2+2z+1}{501z^2-798z+301}$$

a2) $H(z)$ – metoda matched (alternativă)

$$p_d = e^{-p_e T_e}, \quad z_d = e^{-z_e T_e} \quad (\text{pas 1: poli \& zerouri})$$

dacă $n > m$: se adaugă $(n-1)$ factori $(z+1)$ la numărător (pas 2)

$$\lim_{s \rightarrow 0} s^\sigma G(s) = \lim_{z \rightarrow 1} \left(\frac{z-1}{zT_e} \right)^\sigma G(z) \quad (\text{pas 3: câștigul } K_d, \sigma = \text{tipul sistemului})$$

Poli $G(s)$: $s=0 \rightarrow z=1$; $s=-5 \rightarrow z=e^{-5 \cdot 0.1} = e^{-0.5} = 0.6065$. Fără zerouri finite ($m=0$), $n=2 \Rightarrow (n-1)=1$ factor $(z+1)$ adăugat:

$$G(z) = \frac{K_d(z+1)}{(z-1)(z-0.6065)}$$

Sistem tip 1 ($\sigma=1$):

$$\lim_{s \rightarrow 0} sG(s) = \lim_{s \rightarrow 0} \frac{1}{s+5} = \frac{1}{5} \quad \lim_{z \rightarrow 1} \frac{z-1}{zT_e} \cdot \frac{K_d(z+1)}{(z-1)(z-0.6065)} = \frac{2K_d}{T_e(1-0.6065)}$$

$$\frac{1}{5} = \frac{2K_d}{0.1 \cdot 0.3935} \Rightarrow K_d = 0.0039$$

$$G(z) = \frac{0.0039(z+1)}{(z-1)(z-0.6065)} = \frac{0.0039z+0.0039}{z^2-1.6065z+0.6065}$$

$$H(z) = \frac{G(z)}{1+G(z)} = \frac{0.0039z+0.0039}{z^2-1.6026z+0.6105}$$

(comparativ cu Tustin: aceiași pași b)/c) se aplică identic, doar cu polii/coeficienții de mai sus.)

b) Eroare staționară

$$G(z) = \frac{(z+1)^2}{100(z-1)(5z-3)}$$

$$K_p = \lim_{z \rightarrow 1} G(z) = \infty \quad K_v = \frac{1}{T_e} \lim_{z \rightarrow 1} (z-1)G(z) = \frac{1}{0.1} \cdot \frac{4}{100 \cdot 2} = 10 \cdot 0.02 = 0.2$$

$$K_a = \frac{1}{T_e^2} \lim_{z \rightarrow 1} (z-1)^2 G(z) = \lim_{z \rightarrow 1} (z-1) \cdot \frac{(z+1)^2}{T_e \cdot 100(5z-3)} = 0$$

$$e_{ss}(\text{treaptă}) = \frac{1}{1+K_p} = 0 \quad e_{ss}(\text{rampă}) = \frac{1}{K_v} = \frac{1}{0.2} = 5 \quad e_{ss}(\text{parabolă}) = \frac{1}{K_a} = \infty$$

c) Răspuns la impuls unitar Dirac ($\mathcal{Z}^{-1}$)

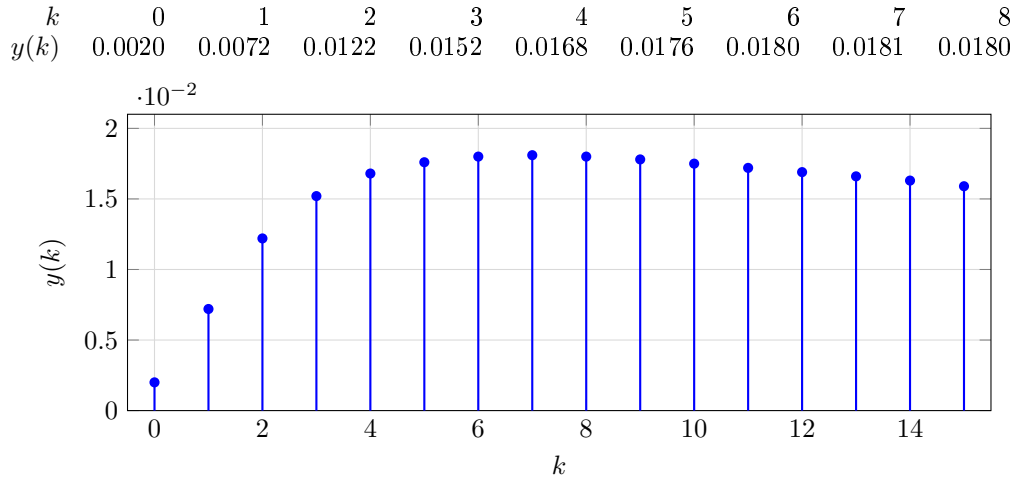
$$Y(z) = H(z) \cdot R(z) = H(z) = \frac{z^2+2z+1}{501z^2-798z+301}$$

$$z_{1,2} = \frac{798 \pm \sqrt{798^2 - 4 \cdot 501 \cdot 301}}{2 \cdot 501} \Rightarrow z_1 = 0.9793, \quad z_2 = 0.6135$$

$$\frac{Y(z)}{z} = \frac{C_0}{z} + \frac{C_1}{z-z_1} + \frac{C_2}{z-z_2} \quad \mathcal{Z}^{-1}\left\{\frac{z}{z}\right\} = \delta(k), \quad \mathcal{Z}^{-1}\left\{\frac{z}{z-a}\right\} = a^k$$

$$C_0 = \frac{1}{301} = 0.0033 \quad C_1 = \frac{(z_1+1)^2}{501 z_1(z_1-z_2)} = 0.0218 \quad C_2 = \frac{(z_2+1)^2}{501 z_2(z_2-z_1)} = -0.0232$$

$$y(k) = 0.0033 \delta(k) + 0.0218 (0.9793)^k - 0.0232 (0.6135)^k, \quad k \geq 0$$



2. $G_d(z) = \frac{0.2z + 0.07}{z^2 - 0.5z + 0.05}$, $T_e = 1s$ (circuit deschis)

$$z = \frac{1 + \frac{T_e}{2}w}{1 - \frac{T_e}{2}w} \quad T_e=1 \quad \frac{1 + 0.5w}{1 - 0.5w} = \frac{N}{D}, \quad N = 1 + 0.5w, \quad D = 1 - 0.5w$$

$$0.2z + 0.07 = \frac{0.2N + 0.07D}{D} = \frac{0.27 + 0.07w}{D}$$

$$z^2 - 0.5z + 0.05 = \frac{N^2 - 0.5ND + 0.05D^2}{D^2} = \frac{0.39w^2 + 0.95w + 0.55}{D^2}$$

$$G_d(w) = \frac{(0.27 + 0.07w)D}{0.39w^2 + 0.95w + 0.55} = \frac{-0.03w^2 - 0.07w + 0.27}{0.39w^2 + 0.95w + 0.55}$$

$$N(w) = 0 \Rightarrow w_{z1} = 2, \quad w_{z2} = -4.15 \quad D(w) = 0 \Rightarrow w_{p1} = -0.94, \quad w_{p2} = -1.51$$

$$G_d(w) = \underbrace{\frac{0.27}{0.55}}_{K_B=0.49} \cdot \frac{\left(1 - \frac{w}{2}\right) \left(1 + \frac{w}{4.15}\right)}{\left(1 + \frac{w}{0.94}\right) \left(1 + \frac{w}{1.51}\right)}$$

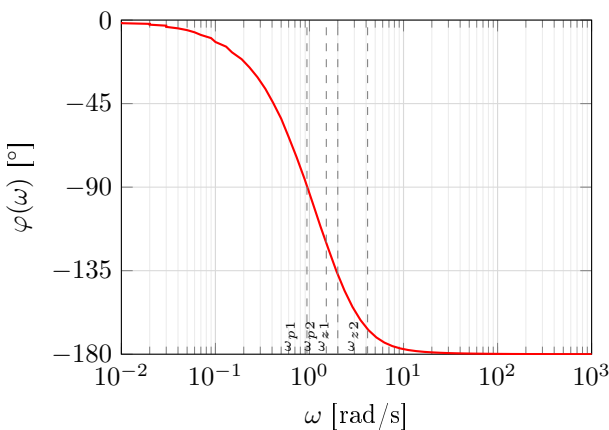
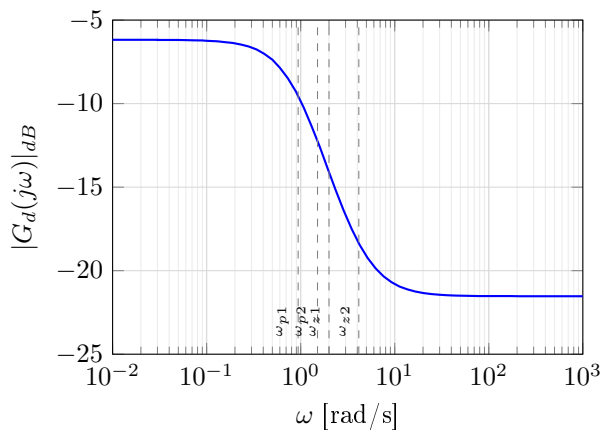
$w_{z1} = 2 > 0 \Rightarrow$ zerou neminim-fazic. Frecvențe de frângere (rad/s): $\omega_{p1} = 0.94$, $\omega_{p2} = 1.51$, $\omega_{z1} = 2$, $\omega_{z2} = 4.15$.

Panta asimptotică a modului (nivel de start $20 \lg K_B = -6.18\text{dB}$):

$$\underbrace{0}_{\omega < 0.94} \rightarrow \underbrace{-20}_{0.94 < \omega < 1.51} \rightarrow \underbrace{-40}_{1.51 < \omega < 2} \rightarrow \underbrace{-20}_{2 < \omega < 4.15} \rightarrow \underbrace{0}_{\omega > 4.15} \text{ dB/dec}$$

Nivel final ($\omega \rightarrow \infty$): $20 \lg \left| \frac{-0.03}{0.39} \right| = -21.53\text{dB}$ (raport coeficienți dominanți). Fază asimptotică ($\omega \rightarrow \infty$): $2 \text{ poli}(-90^\circ) + 1 \text{ zerou m.f.}(+90^\circ) + 1 \text{ zerou n.m.f.}(-90^\circ) = -180^\circ$.

ω	0.01	0.1	0.5	1	2	5	10	50	100
$ G_d _{dB}$	-6.18	-6.24	-7.39	-9.84	-14.09	-19.14	-20.80	-21.50	-21.52
$\varphi [^\circ]$	-1.1	-11.4	-53.5	-93.3	-137.0	-170.5	-177.3	-179.7	-179.8



$$3. G_d(z) = \frac{k(0.13z + 0.02)}{z^2 - 0.39z + 0.007}, T_e = 1s$$

$$z^2 - 0.39z + 0.007 = 0 \Rightarrow p_1 = 0.37, p_2 = 0.02 \quad 0.13z + 0.02 = 0 \Rightarrow z_0 = -0.15$$

a) Ramuri

$n = 2, m = 1 \Rightarrow$ nr. ramuri = 2.

Ramuri din $p_1 = 0.37, p_2 = 0.02$; se termină în $z_0 = -0.15$ și în $z \rightarrow \infty$.

b) Axa reală

$$(p_2, p_1) = (0.02, 0.37) \quad (-\infty, z_0) = (-\infty, -0.15)$$

c) Asimptote

$$n - m = 1, \quad \theta = \frac{180^\circ(2i + 1)}{n - m} \Big|_{i=0} = 180^\circ \quad \sigma_a = \frac{\sum p_i - \sum z_i}{n - m} = \frac{0.39 - (-0.15)}{1} = 0.54$$

1 asimptotă, $180^\circ, \sigma_a = 0.54$.

d) Puncte de ramificație

$$1 + G_d(z) = 0 \Rightarrow k(z) = -\frac{N(z)}{D(z)}$$

$$N(z) = z^2 - 0.39z + 0.007, \quad D(z) = 0.13z + 0.02$$

$$\frac{dk}{dz} = 0 \Leftrightarrow N'(z)D(z) - N(z)D'(z) = 0, \quad N' = 2z - 0.39, D' = 0.13$$

$$(2z - 0.39)(0.13z + 0.02) - 0.13(z^2 - 0.39z + 0.007) = 0$$

$$0.13z^2 + 0.04z - 0.01 = 0$$

$$z = \frac{-0.04 \pm \sqrt{0.04^2 + 4 \cdot 0.13 \cdot 0.01}}{2 \cdot 0.13}$$

$$z_{b1} = 0.15 \quad (k \approx 0.73) \quad z_{b2} = -0.45 \quad (k \approx 10.00)$$

e) Cerc & trasare

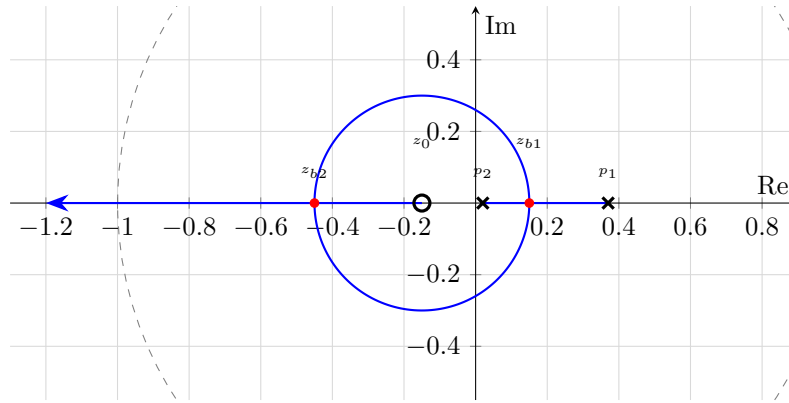
$$z = z_0 + w, \quad |w|^2 = (z_0 - p_1)(z_0 - p_2)$$

$$r = |w| = \sqrt{(z_0 - p_1)(z_0 - p_2)}$$

$$z_0 - p_1 = -0.52 \quad z_0 - p_2 = -0.17 \quad r = \sqrt{0.09}$$

$$r = 0.30$$

$$|z_{b1} - z_0| = |0.15 - (-0.15)| = 0.30; |z_{b2} - z_0| = |-0.45 - (-0.15)| = 0.30 \checkmark$$



(cerc punctat = cerc unitate.)

f) Curbă polară ($k = 1$)

$$z = \frac{1 + \frac{T_e}{2}w}{1 - \frac{T_e}{2}w} \quad T_e = 1 \quad \frac{1 + 0.5w}{1 - 0.5w} = \frac{N}{D}, \quad N = 1 + 0.5w, \quad D = 1 - 0.5w$$

Se înlocuiește $z = N/D$ în $G_d(z)$ (ca la Bode) și se amplifică cu D^2 :

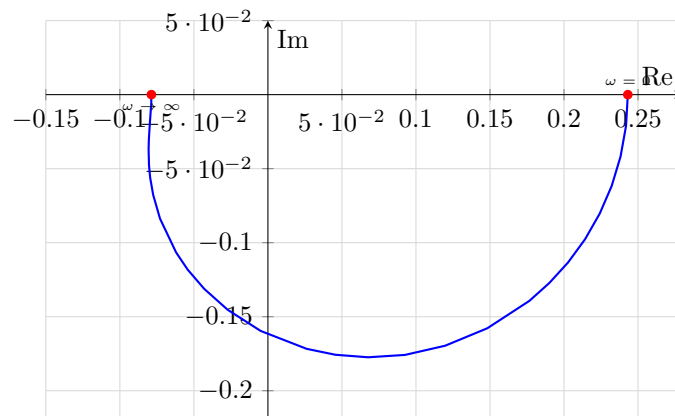
$$G_d(w) = \frac{(0.13N + 0.02D)D}{N^2 - 0.39ND + 0.007D^2} = \frac{-0.0275w^2 - 0.02w + 0.15}{0.34925w^2 + 0.993w + 0.617}$$

$$w = j\omega \Rightarrow G_d(j\omega) = \text{Re}(\omega) + j \text{Im}(\omega), \quad \omega \in [0, \infty)$$

Capetele curbei: $\omega = 0$ ($z = 1$); $\omega \rightarrow \infty$ ($z = -1$).

$$G_d(j0) = \frac{0.15}{0.617} = 0.2431 \text{ (start)} \quad \lim_{\omega \rightarrow \infty} G_d(j\omega) = \frac{-0.0275}{0.34925} = -0.0787 \text{ (final)}$$

ω	0	0.2	0.5	1	2	3	5	10	30
Re	0.2431	0.2241	0.1482	0.0262	-0.0620	-0.0775	-0.0806	-0.0796	-0.0788
Im	0	-0.0804	-0.1578	-0.1717	-0.1066	-0.0677	-0.0370	-0.0172	-0.0056



$$4. G_0(z) = \frac{0.02z + 0.02}{z^2 - 1.49z + 0.54}$$

a) Forma complet controlabilă

$$G_0(z) = \frac{b_1z + b_0}{z^2 + a_1z + a_0}, \quad a_1 = -1.49, \quad a_0 = 0.54, \quad b_1 = b_0 = 0.02$$

$$y(k+2) + a_1y(k+1) + a_0y(k) = b_1r(k+1) + b_0r(k)$$

$$x_1(k) = y(k), \quad x_2(k) = x_1(k+1) - b_0r(k)$$

$$x_1(k+1) = x_2(k) \quad x_2(k+1) = -a_0x_1(k) - a_1x_2(k) + r(k) \quad y(k) = b_0x_1(k) + b_1x_2(k)$$

$$\Phi = \begin{pmatrix} 0 & 1 \\ -0.54 & 1.49 \end{pmatrix}, \quad \Gamma = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad C = (0.02 \quad 0.02), \quad D = 0$$

$$zI - \Phi = \begin{pmatrix} z & -1 \\ 0.54 & z - 1.49 \end{pmatrix}, \quad \det = z^2 - 1.49z + 0.54$$

$$C(zI - \Phi)^{-1}\Gamma = \frac{(0.02 \quad 0.02) \begin{pmatrix} 1 \\ z \end{pmatrix}}{z^2 - 1.49z + 0.54} = \frac{0.02z + 0.02}{z^2 - 1.49z + 0.54} = G_0(z) \quad \checkmark$$

b) Observabilitate

$$Q = \begin{pmatrix} C \\ C\Phi \end{pmatrix}, \quad C\Phi = (0.02 \quad 0.02) \begin{pmatrix} 0 & 1 \\ -0.54 & 1.49 \end{pmatrix} = (-0.0108 \quad 0.0498)$$

$$Q = \begin{pmatrix} 0.02 & 0.02 \\ -0.0108 & 0.0498 \end{pmatrix} \quad \det Q = 0.02 \cdot 0.0498 - 0.02 \cdot (-0.0108) = 0.0012 \neq 0$$

$\text{rang}(Q) = 2 = n \Rightarrow$ **sistem complet observabil.**